CS612 Lecture 02 — In-Class Exercises

Divide-and-Conquer, Recurrences & Selection · Sat 03 Oct 2026 · ungraded

Opening

What is the best lower bound you can prove, right now, for finding the median of n distinct keys in the comparison model? Why does the decision-tree argument from L01 give almost nothing here?

A — Classify and solve

For each recurrence: does the Master theorem apply? If so, which case (check regularity where needed)? If not, which tool would you use? Give the Θ bound.

1. $T(n) = 8T(n/2) + n^2$
2. $T(n) = 2T(n/4) + \sqrt{n}$
3. $T(n) = 2T(n/2) + n/\log n$
4. $T(n) = 3T(n/3) + n\log^2 n$
5. $T(n) = T(n/2) + T(n/4) + n$
6. $T(n) = 2T(n/2) + n^2$

A2 — Substitution practice (if time permits)

Prove by substitution that $T(n) = 4T(n/2) + n$ satisfies $T(n) = O(n^2)$. First try the hypothesis $T(n) \le cn^2$ and explain exactly where it fails. Then strengthen it.

B — Selection

1. In the median-of-medians lemma, why do we exclude "the group containing x and a possibly short last group"? What is the exact bound on $|L|$ if n is a multiple of 10?
2. Redo the analysis for groups of **7**: give the size bound for the larger side and the recurrence, and prove $T(n) = O(n)$ by substitution with an explicit constant.
3. Kadane's algorithm: state the invariant and prove the maintenance step. Modify the algorithm to return the indices of an optimal subarray.

C — Pick the method (and defend it)

For each scenario give the method, its cost *with the guarantee type*, and the assumption that, if it changed, would change your answer.

1. A public web API returns the median of a client-supplied array ($n \le 10^6$) under a strict latency SLO. Attackers may submit crafted arrays.
2. A stream of 10^{10} events; report the 100 largest values at the end. Memory: 1 GB.
3. A fixed data set of $n = 10^8$ values will receive 10^5 rank queries with different k over the next hour.
4. An in-memory array of $n = 10^9$ values; one report of the 1st, 2nd, …, 99th percentiles.

Take-home experiment (optional)

Run `Student_Resources/selection_bench.cpp`. (1) Explain why the quickselect comparison count per element stays near 3.4. (2) Construct an input that makes a *deterministic first-element-pivot* quickselect quadratic, and measure it. (3) Try to construct an input on which median-of-medians with groups of 3 shows clear $n \log n$ growth. Is it easy? Why or why not?